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In it to Win it: Competitiveness, Concussion Knowledge and Nondisclosure in Athletes

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In it to win it: Competitiveness, concussion knowledge and nondisclosure in athletes

Objectives: Despite having a general understanding of concussions, many athletes choose not to report symptoms of a sports-related concussion (SRC) which leads to a larger burden on our healthcare system due to longer recoveries, more Emergency Department visits, and future medical appointments. Thus, there is a need to identify factors, such as competitiveness, that may help to better explain the nondisclosure of concussion symptoms. The current study aimed to investigate the role of athletes' concussion knowledge and competitiveness on concussion nondisclosure. **Methods:** Participants included 161 Canadian athletes (ages 14-32; 71% male) participating in collision sports (American football, rugby, hockey), soccer or rowing. Concussion knowledge, reporting and competitiveness were measured using the Rosenbaum Concussion Attitude and Knowledge Survey and Sports Orientation Questionnaire (SOQ). Two logistic regression models were performed to explain both history and future intention of playing despite experiencing concussion symptoms. **Results:** Age, sex, and concussion knowledge were not significant for the regression models. The SOQ scores (measuring competitiveness) were significant for both regression models, such that higher competitiveness was associated with increased likelihood for past nondisclosure (b=.03, p=.03, OR=1.03 (97.5% CI: 1.003,1.06)) and future nondisclosure intention (b=.05, p=.00, OR=1.05 (97.5% CI: 1.03,1.08)). Regarding concussion knowledge, soccer players had significantly higher knowledge than those in collision sports and rugby, $F(2, 158)= 140.5$, p=.00; female athletes had significantly higher knowledge than males, $t(66)=-4.26$, p=.00; age was negatively associated with concussion knowledge (r=-0.35, p<.01). **Conclusion**: The results suggest that healthcare providers should be aware that athletes tend to have adequate concussion knowledge, but this knowledge does not explain past nondisclosure or future nondisclosure intention. Based on these findings, healthcare providers should focus on being able to identify highly competitive athletes who are most at risk of nondisclosure instead of exclusively aiming to increase concussion knowledge in athletes.

Keywords*:* head injuries; brain concussion; traumatic brain injury; self-report, sports psychology

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Introduction

In the past five years, the number of emergency department visits for sports-related brain injuries in Ontario and Alberta has increased by 15%, with over 16,000 visits reported in 2017 - 2018. The vast majority (93%) of the brain injuries were diagnosed as sports-related concussions [1]. Sports-related concussion (SRC) is characterized as "a traumatic brain injury induced from biomechanical forces" such as contact with another athlete, equipment, or playing surface [2, p.839]. After experiencing SRC, most athletes completely recover within 7 - 10 days [3], particularly when the appropriate return-to-play protocol is followed [4,5]. However, there is a vulnerable period in recovery when a second impact can result in intracerebral hemorrhage, cerebral edema, or death [6-8]. Thus, athletes must have adequate concussion knowledge so that they can immediately identify any potential concussion symptoms. Continuing to play while symptomatic can not only prolong recovery but may have life-threatening consequences that unnecessarily increase the burden on our healthcare system.

Concussion Knowledge

Past research on SRC knowledge is often limited to either high school or collegiate varsity athletes. Knowledge is usually assessed using survey measures such as the Rosenbaum Concussion Attitude and Knowledge Survey (RoCKAS) [9]. Studies indicate that high school and collegiate athletes are generally knowledgeable about the most common symptoms of SRC such as headaches, dizziness, and confusion [10,11]. Further, previous research has found that both female athletes and older athletes tend to have higher concussion knowledge in comparison to their male and younger counterparts [12-15]. Given the amount of evidence demonstrating that both high school and female athletes are the most vulnerable groups to the aftermath of concussions, the fact they are knowledgeable about SRCs and their effects is promising [3]. Due to the likely absence of healthcare professionals during training, athletes often have to resort to self-diagnosis or coaches' diagnoses; however, there are still concerns that having some knowledge may not be adequate for proper concussion identification. A recent study identified that only 5.7% of combat sport coaches properly recognized the level of traumatic brain injury of an SRC and 68.8% were unfamiliar with any sideline assessment tools [16]. Relatedly, researchers have emphasized the need for concussion education initiatives aimed at younger cohorts of athletes, as well as programs targeted at all levels of sport including family members, coaches, and support staff to maximize acquired concussion knowledge [17,18]. However, studies have also found that having moderate to high knowledge does not predict concussion reporting [19-21]. Thus, despite having a general knowledge of concussion symptoms and understanding the high risk for long-term sequelae, many athletes still choose not to report symptoms of a concussion.

Barriers to Concussion Reporting

In recent years, many studies have attempted to explain why up to 50% of SRC are

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unreported [4,11,22]. It may be the result of a combination of previously identified factors including lack of support, limited coach approachability, reporting skill, and poor organizational safety climate [21,23,24]. Further, a recent systematic review of concussion reporting barriers identified fear of losing current or future playing time as one of the major barriers, followed by the misconception that SRC is not serious, and a fear of letting the team down [25]. Relatedly, Register-Mihalik and colleagues [26] found that many athletes who have already experienced SRC often have lower intentions to report future concussion symptoms due to having reported concussions in the past without experiencing any negative health outcomes and only experiencing negative outcomes in the form of losing playing time. O'Connor and colleagues [19] found similar results in which concussion knowledge, gender and pressure to play did not predict concussion nondisclosure, but those with a history of SRC were significantly more likely to not report an SRC. Previous research has also investigated the prevalence of concussion nondisclosure in collision (American football), medium/high contact (soccer, basketball, lacrosse, etc.), and low/no contact sports (baseball, rowing, golf, etc.) and found that nondisclosure rates were significantly higher in collision sports in comparison to medium/high or low/no contact sports [27]. Similarly, Kuhn and colleagues [28] determined that collision (American football) athletes were significantly more likely to play through a concussion in comparison to athletes playing full/medium contact or no/low contact sports after analyzing 266 middle school, high school, and collegiate athletes. While it seems that researchers have identified many barriers to reporting, none have been able to clearly explain why an athlete decides not to report a concussion.

Competitiveness as a Barrier to Reporting

Self-Determination Theory states that human behaviour is motivated by three needs:

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autonomy, competence, and relatedness, with the key outcome being self-determination through intrinsic motivation [29]. Self-Determination Theory posits that external factors such as winning, competition, and rewards result in lower intrinsic motivation, in what is known as the "over-justification" effect [30]. An athlete is intrinsically motivated when they freely engage in their sport because they find it stimulating, gratifying and an opportunity to learn while athletes who engage in sports because they value the associated outcomes such as reward and recognition are extrinsically motivated [31]. In fact, competitive athletes tend to focus on the outcome of winning, which is entirely extrinsically motivated. However, high competitiveness is not inherently a negative trait and is often seen as a desirable trait in an athlete. Highly competitive athletes accept challenges, strive for high-performance standards, and persevere until they attain their goals [32]. Perhaps then, highly competitive and subsequently, extrinsically motivated athletes are less likely to report a concussion because it may interfere with their efforts to succeed and meet their goals, and ultimately, does not reinforce the external motivation of winning.

Competitiveness, defined as "the desire to win in interpersonal situations" [33,p. 290], is seen as a personality trait that is not often prone to malleability [34]. It is most commonly assessed using questionnaires such as the Sport Orientation Questionnaire (SOQ; [34]). The SOQ has been favoured in research with athletes because it is sports-focused, has good reliability indicators and is suitable for a comprehensive range of ages [33,34]. Gill & Dzewaltowski [35] found that athletes are more achievement-oriented and competitive than non-athletes and that athletes obtained significantly higher scores than non-athletes on the SOQ. So, if an athlete has an extreme desire to win and succeed, they will consequently not want to lose current or future playing time by reporting a potential concussion and being unable to play. While having adequate

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concussion knowledge is important, an athlete's competitiveness may be the crucial factor that disrupts the link between recognizing the symptoms of a concussion and reporting those symptoms. Being able to identify traits of athletes least likely to report SRC, such as those with higher levels of competitiveness, can help coaches, parents, and teammates detect players most at risk of continuing to play with an SRC, and do so more concretely than previously identified barriers.

Present Study

There is a need to explore SRC knowledge and nondisclosure in a larger age range and more diverse sample of athletes playing sports at different contact levels who may have less access to resources to learn about SRC. Thus, the overall objective of the study is to explore SRC knowledge and various potential factors associated with nondisclosure including competitiveness, age, sex, and sport in a diverse sample of adolescent and young adult athletes playing a wide range of sports (i.e., football, hockey, rugby, soccer and rowing). We hypothesized that concussion knowledge will be highest in the collision sports group due to higher exposure to SRC as well as more opportunities for education. We also hypothesized that higher competitiveness levels would be associated with a higher likelihood of nondisclosure in all athletes regardless of sport, sex, or age.

Materials and Methods

Participants

This study utilized data collected from adolescent and young adult participants who play at an elite level for their local community, high school or university sports team from a mid-sized Canadian city. The collision sports (American football, rugby, and hockey) were grouped together while rowing (a noncontact sport) and soccer (a high/medium contact sport) were separated using previous research for contact classification [27,28,36,37]. The original design aimed to have two sports for each

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contact level with lacrosse and soccer for high/medium contact sports, and swimming and rowing for noncontact sports. However, we could not find any swimming or lacrosse teams willing to participate in the current study at the time of data collection; therefore, our sample consisted of athletes from three sports in collision sports as well as soccer and rowing. Athletes from all ethnicities, genders, and socioeconomic statuses were welcome to participate in the study as well as those with a positive history of neurological or mental health issues.

Measures

For this cross-sectional study, a survey method was used that included three components: a demographic questionnaire, a concussion knowledge and attitudes questionnaire, and a competitiveness questionnaire. The survey included a modified version of the open-access Rosenbaum Concussion Attitude and Knowledge Survey (RoCKAS) – Supplemental [9] for all demographic information. The modified version included items about sex, gender, other sports activity, education, parental education, concussion questions, and a concussion ranking question that asked participants to rank the order of importance of each decision-maker on an athlete's return to play following a concussion. We excluded questions from the supplemental survey on high school information as it was not relevant to all participants and used American language such as "Freshman" and "Varsity" that may be confusing to Canadian participants.

Concussion Attitudes and Knowledge.

The open-access Rosenbaum Concussion Attitude and Knowledge Survey - Student version (RoCKAS-ST) was given to participants to gain information on attitudes towards concussions and knowledge of concussions [9]. The RoCKAS-ST contains a 15-item Concussion Attitudes Index (CAI). All 15 items are statements on concussion attitudes that participants rated on a Likert scale of 1 (Strongly disagree) to

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5 (Strongly agree). An example of an item is "I feel that concussions are less important than other injuries". The CAI ranges from 15 – 75, with higher scores indicating safer attitudes towards concussions.

The RoCKAS-ST also contains 17 true or false items and a list of 16 potential concussion symptoms, eight of which are distractor items (i.e., hair loss, excessive studying) that comprise the Concussion Knowledge Index (CKI). The CKI scoring was modified from a max of 8 points for potential concussion symptoms to 16 points in such a way that if the participant did not mark the distractor, they received one point for accuracy and if they did mark the distractor, they got a score of zero for that item. With this modification, the CKI has a maximum score of 33, with higher scores indicating greater concussion knowledge.

Competitiveness

Levels of competitiveness were assessed in each participant using the 25-item Sport Orientation Questionnaire (SOQ), a reliable and valid measure of sports achievement orientation that includes items that comprise three factors: goal orientation (e.g., "I am most competitive when I try to achieve personal goals", win orientation (e.g., "I have the most fun when I win"), and competitiveness (e.g., "I thrive on competition") [34]. All 25 items are statements that participants rate from 1 (Strongly Disagree) to 5 (Strongly Agree). The SOQ has a maximum score of 125, with higher scores indicating a higher level of competitiveness.

Concussion nondisclosure

Concussion nondisclosure was assessed using two items. The first item was a dichotomous (Yes/No) item from the RoCKAS Supplemental survey and reads, "Have you ever received what you thought was a concussion and kept playing without telling anyone about it?". By continuing to play with what the individual believed was a

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concussion, athletes are ultimately choosing to not report a potential concussion. Thus, it can be used as a measure of past concussion nondisclosure.

The second item was from Rosenbaum and Arnett's CKI which is rated on a 5-point-Likert scale that states "I would continue playing a sport while also having a headache that resulted from a minor concussion," with higher scores indicating greater agreement. By continuing to play with a symptom of a concussion, athletes are choosing to not report the symptom. Since it reflects the overall appraisal of continuing to play while symptomatic, it can be considered a face valid proxy for concussion nondisclosure intention [38].

Procedure

Recruitment of participants was initiated by e-mailing coaches of local sports teams. If coaches expressed interest in having their athletes participate in the study, a Letter of Information for Implied Consent was sent to the coaches to distribute to the players and/or parents (if under 18 years old). The coaches were then contacted to discuss a suitable time for data collection, which was either before or after a scheduled practice. During the visit, a second copy of the Letter of Information for Implied Consent was distributed to the players for review. Once the participants reviewed the letter, the questionnaires were distributed and anonymously completed. For each team, all athletes completed the questionnaire simultaneously and independently under the supervision of a minimum of three research assistants, with no access to internet or other educational materials. The athletes were not compensated for their participation in the study. The study was approved by the local University's Human Research Ethics Board (protocol number: 18-1106).

Analysis

Descriptive univariate statistics for sample characteristics, concussion

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knowledge, attitudes and competitiveness were computed. Pearson's correlation was conducted to assess potential relationships between knowledge, attitudes, competitiveness, and age. Normality was assessed using Shapiro-Wilk's test, while homogeneity of variance was assessed using Levene's test. T-tests were used to detect potential sex differences in knowledge, attitudes and competitiveness. Similarly, ANOVA and Tukey's HSD post-hoc analysis were utilized to detect any differences between rowing, soccer, and collision sports in knowledge, attitudes, and competitiveness. Ordinal logistic regression was conducted to assess the association between competitiveness and concussion nondisclosure intention. Additionally, a binomial logistic regression was performed to assess the association between competitiveness and past concussion nondisclosure. In both analyses, concussion knowledge, sex, sport, and age were included as additional variables in the model. Three participants were not included in the first logistic regression analysis as they did not answer the past nondisclosure item and could not be included in the model; similarly, two participants were not included in both logistic regression analyses as they did not report their age. The resulting odds ratios (ORs) were presented with 97.5% confidence intervals (CI). An alpha level of $\leq$ .05 was used as a cutoff for statistical significance. Analyses were completed using the computing environment 'R Studio' version 1.2.1335 [39].

Results

Table 1 details the demographic characteristics of the sample, distributed by sport (i.e., rowing, soccer, collision sports). There was a total of 191 participants, but 16% (n = 30) were removed due to incomplete data (i.e., missing one or more items needed to compute a total score for concussion knowledge or competitiveness). Of the included 161 participants, 71.1% were male and were an average age of 17.6 years (*SD*

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= 3.95; ranging from 14 to 32 years of age). The majority (77%) of the players identified as Caucasian, and, there was also a representation of different ethnicities (e.g., Indigenous, Asian).

Concussion attitudes and knowledge

The average CAI score for all participants was 59.2 (*SD* = 7.85) out of a maximum score of 75. There were no significant differences in attitudes based on sex or sport. There were also no significant correlations between CAI and CKI, SOQ or age (see Table 2 [40]).

The average CKI score for all participants was 20.06 (*SD* = 3.47) out of a maximum score of 33. However, there were significant differences in concussion knowledge between males (*M* = 19.28, *SD* = 2.93) and females (*M* = 22.04, *SD* = 3.98), with female athletes having significantly higher knowledge scores, $t(66) = -4.26$, $p = .00$. Further, an ANOVA indicated that there were significant differences in concussion knowledge between sports, $F(2, 158) = 140.5$, $p = .00$, with the soccer athletes (*M* = 25.32, *SD* = 2.49) having significantly higher knowledge scores than both the rowing (*M* = 18.64, *SD* = 1.65) and collision groups (*M* = 18.60, *SD* = 2.03). Interestingly, there were no significant differences in concussion knowledge between the rowing and the collision sports groups. There was a significant negative correlation ($r = -0.35$, $p < .01$) between CKI and age, such that higher knowledge was associated with younger participants (see Table 2). The mean scores for CKI and CAI across sex and sport are shown in Figure 1.

Competitiveness

Using Cronbach's α, internal consistency reliability of the SOQ scores was estimated to be .95. The average SOQ score was 104.75 (*SD* = 14.29) out of a maximum score of 125 and was slightly negatively skewed (-.59). There were no

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significant differences in competitiveness levels between the three sports groups. There were significant differences in levels of competitiveness between sexes, $t(109) = 2.81$, $p = .00$. Male athletes ($M = 106.7$, $SD = 14.88$) had significantly higher competitiveness scores than females ($M = 100.59$, $SD = 11.31$). There was also a significant positive correlation ($r = 0.24$, $p < .01$) between competitiveness and age, such that higher competitiveness levels were associated with older participants.

Concussion nondisclosure

Almost half of the athletes (44%) reported that they had received what they thought was a concussion and kept playing without telling anyone about it. Further, the average score from 1 (Strongly disagree) to 5 (Strongly agree) for concussion reporting intention was 3.06 ($SD = 1.34$).

The ORs for each model are presented in Table 3. Age, sex, and concussion knowledge were not significant in either model. The first model shows the significant effects of competitiveness ($b = .03$, $p = .03$) and rowing ($b = -2.05$, $p = .01$) on past concussion nondisclosure. In this model, 18.5% (Nagelkerke R^2) of variance is accounted for. As the SOQ scores increased by one unit, the change in the odds of past nondisclosure (rather than no history of nondisclosure) is 1.03. Thus, those with higher competitiveness levels are more likely to have not disclosed a concussion in the past. Lastly, and as expected, rowing athletes are 7.78 times more likely than collision sports athletes to have no history of concussion nondisclosure.

The second model shows a significant effect of competitiveness ($b = .05$, $p = .00$) on concussion nondisclosure intention with an OR of 1.05 (97.5% CI: 1.003, 1.06). In other words, as competitiveness increases, concussion nondisclosure intention also increases. In this model, 18.4% (Nagelkerke R^2) of variance is accounted for.

Discussion

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While there have been previous studies investigating concussion knowledge and nondisclosure, our study is one of the first to explore knowledge and nondisclosure across a cohort of athletes from adolescence to young adulthood participating in a wide range of sports of varying levels of contact. Although past research has stated that athletes may not report concussive symptoms due to lack of knowledge of symptoms, we found that concussion knowledge was not associated with future reporting intention or past nondisclosure. This study highlights that competitiveness may be a barrier to reporting, regardless of age, sex, and sport. These findings regarding nondisclosure may help parents, coaches and healthcare providers seeking to improve concussion reporting by acknowledging that certain athletes (i.e., extremely competitive) are more at risk and should be specifically targeted during interventions.

Concussion knowledge

The findings of our study support some of the previous findings on SRC knowledge in athletes, but also provides new evidence that has yet to be considered concerning athletes' SRC knowledge. We found a statistically significant difference between male and female athletes in concussion knowledge that is consistent with previous findings that female athletes have greater knowledge of SRC in comparison to male athletes [12-14]. However, similar to the findings by Wallace and colleagues [41], the difference between knowledge scores in our study was small and our results may not demonstrate a clinically meaningful difference in knowledge between the two sexes, particularly in the context of the unequal distribution of sex across sports. Therefore, these results should be interpreted with caution. Also, our study indicated that younger athletes had higher knowledge. This is a positive finding, given that adolescents have been identified to be at a higher risk for the detrimental effects of concussions [3]. Our findings that younger athletes have more knowledge seems to be in contrast to previous

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research [13,17]; however, there has been a fairly recent push for more concussion education initiatives in youth sports. Thus, older athletes may have less knowledge considering concussion awareness and education may not have been as prevalent in their youth and they have consequently had fewer opportunities for education. Lastly, there were significant differences in knowledge between sports, with soccer players having higher knowledge than both the rowing and collision groups. We predicted that collision sports would have the highest knowledge due to more exposure to concussions and education efforts, but our findings did not support this hypothesis. However, the difference in knowledge, while statistically significant, was small and may not be clinically significant, particularly considering the unequal sample sizes across sports groups and should be interpreted with caution. Nanos and colleagues [15] identified no significant differences in knowledge between contact and no contact sports, even though the contact sports group had higher perceived knowledge and were more likely to recognize that an athlete needs to be symptom-free before returning to play. Further, Beidler and colleagues [20] found that there were no differences in SRC knowledge between traditional (e.g., football, hockey, soccer) and non-traditional (e.g., rugby, boxing, fencing) sports. Considering this is one of the first studies to compare knowledge across sports with differing levels of contact, future research should investigate this difference more thoroughly to elucidate if it replicates and further, if it has any clinical impact.

Competitiveness

This study confirms previous research on competitiveness in athletes but also expands this research to present a new barrier to concussion reporting that had yet to be investigated. While the competitiveness distribution was slightly negatively skewed, there were no significant differences in competitiveness between the three sports

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groups. We anticipated that the competitiveness distribution would be slightly skewed as it is consistent with previous research that most athletes are competitive and would score high on any competitiveness measures [34]. Further, we did find a significant difference in competitiveness between sexes, such that males had slightly higher competitiveness scores. Although the difference in competitiveness scores were minimal, they were statistically significant and consistent with the extant literature. [35,42-45]. Lastly, there was a small positive correlation between age and competitiveness such that as age increases, competitiveness also increases. Mayr and colleagues [42] found that competitive preferences increase until the age of 50 where it starts to decline, forming an inverted-U trajectory. A possible explanation for this initial age increase is supported by a meta-analysis [46] in which the personality trait social dominance (reflected by qualities like dominance, independence, and self-confidence in a social context) showed the only strong positive trend from early adulthood to the fifties. Social dominance is established by successfully engaging in competition and therefore, it is plausible that as one ages, an increased interest in social dominance also produces an increased level of competitiveness [42].

To our knowledge, we are the first to investigate athletes' levels of competitiveness in the context of SRC nondisclosure. As we predicted, competitiveness was found to be a significantly associated with past concussion nondisclosure and future nondisclosure intention. Our findings give evidence to support two separate pathways between the desire to win and the intent to report a future concussion, as suggested by Kroshus and colleagues [47] who reported a significant negative association between conformity to the masculine norm of winning and continuing to play with concussion symptoms in collegiate athletes. Furthermore, Taylor and Sanner [48] found that symptom reporting behaviours heavily depended on the athletes' current state and

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environment (e.g., the importance of the game being played and how secure the athlete felt in their environment). When considering Taylor and Sanner's [48] findings, our results suggest that the importance of a game may influence highly competitive players more strongly to under-report symptoms of a concussion than non-competitive players. Thus, considering that competitive athletes are often extrinsically motivated and focused on the outcome of winning, they are less likely to report SRC symptoms because it will interfere with their efforts to achieve the desired outcome of winning.

Limitations and Future Research

Firstly, albeit a well-represented sample, this study did not have enough participants to compare across genders or ethnicities. Further, the sample was male-dominated (71%), and a more equal gender distribution would be more representative of today's athletes. While our original design intended to have an equal numbers of sports across three contact levels (collision, high/medium and low/noncontact), we were only able to recruit one high/medium contact sports (i.e., soccer) and one noncontact sport (i.e., rowing) and could not compare across contact levels. Another potential limitation is that since recruitment was facilitated by coaches, the sample could be biased to those with more concussion knowledge considering the coaches may promote more SRC awareness, evidenced by their willingness to participate in the study. Considering each regression model only accounted for approximately 18% of the total variance, there are certainly several other factors that are contributing to athletes' nondisclosure that we did not assess. Potential avenues for further research include exploring how personality and individual differences such as aggression, intrinsic and extrinsic motivation interact with competitiveness and impact one's intention to report a concussion. Lastly, researchers should utilize these findings to create a more effective, targeted psychoeducational program aimed to increase concussion reporting behaviours among

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athletes.

Conclusion

Concussions can have serious consequences for athletes, especially when they are left unmanaged. When an athlete does not report a concussion, they are leaving themselves more vulnerable to further brain injury and have the potential to become a strain on our healthcare system. Thus, we must understand why such a high percentage of athletes do not report concussions. Historically, it has been believed that increasing concussion knowledge would be sufficient to increase athletes' reporting, but researchers have determined that this approach has not been overly impactful. These findings led to the current study which aimed to determine if a high level of competitiveness is a barrier to concussion reporting. While competitiveness was a significant predictor of past nondisclosure and future nondisclosure intention, concussion knowledge was not a significant predictor. Most athletes are knowledgeable about the common symptoms of concussion such as headache and dizziness. Perhaps instead of focusing on solely increasing concussion knowledge, we should also teach physicians, coaches, teammates and parents to be able to identify those most at risk of not reporting concussions (e.g., highly competitive) and then, specifically check up on those athletes when there is a potential SRC. There is then less responsibility on players to report symptoms. Further, physicians and trainers may choose to spend their time talking to athletes about their competitiveness and motivations surrounding nondisclosure, instead of just educating them on concussion symptoms. Overall, this study helped to tease apart the nuanced and complex phenomena of concussion nondisclosure by understanding that for athletes, being highly competitive is a significant and positive predictor of future nondisclosure intention and past concussion nondisclosure.

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Table Captions

Table 1. Demographic information for all participants.

Table 2. Means, standard deviations, ranges, and correlations with confidence intervals of CKI, SOQ, CAI and age.

Table 3. Odds of past concussion nondisclosure and future nondisclosure intention.

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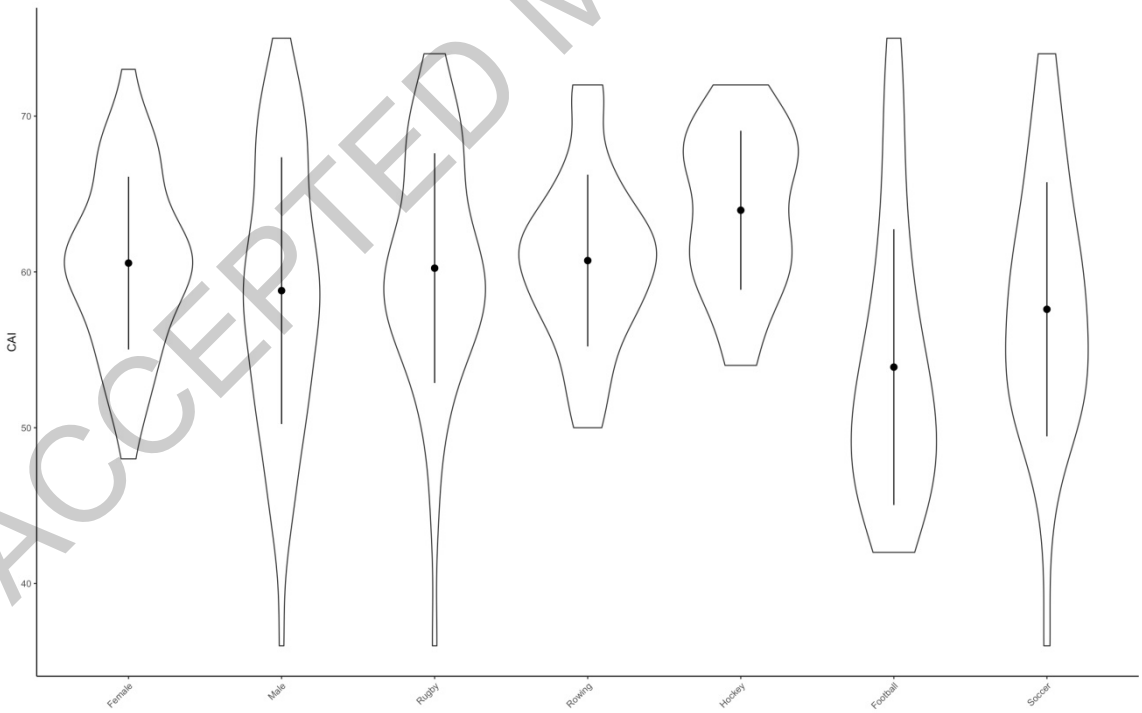

Figure 1

Table 1

Variable	Rowing *n* = 22		Soccer *n* = 35		Collision Sports (American football, rugby and hockey) *n* = 104		Total *n* = 161	
	Mean or Frequency	*SD* or %	Mean or Frequency	*SD* or %	Mean or Frequency	*SD* or %	Mean or Frequency	*SD* or %
Age	15.73	1.08	14.77	.84	18.96	4.31	17.59	3.95
Sex*								
Male	7	31.8 %	14	40.0 %	92	90.2%	113	71.1%
Female	15	68.2 %	21	60.0 %	10	9.8%	46	28.9%
Gender								
Male	6	27.3 %	13	37.1 %	91	89.2%	110	69.2%
Female	15	68.2 %	21	60.0 %	10	9.8%	46	28.9%
Indigenous Two-Spirit	1	4.5%	1	2.9%	1	.98%	3	1.9%
Average Highest Grade Completed	Grade 10		Grade 9		Grade 11		Grade 10	
Years Played	1.58	1.31	8.03	2.69	8.36	5.06		

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Level of Play	High School (100%)		High School (100%)		High School (47%), Men's League (33%), Varsity (19%)		High School (66%), Men's League (22%), Varsity (12%)	
Ethnicity								
Caucasian	10	52.6%	24	75.0%	78	82.1%	112	76.7%
Indigenous	-	-	1	3.1%	3	3.2%	4	2.7%
Asian	5	26.3%	4	12.5%	1	1.1%	10	6.8%
Mixed	4	21.1%	-	-	7	7.4%	11	7.5%
Other	-	-	3	9.4%	6	6.3%	9	6.2%

Notes:. SD = Standard Deviation. *Two participants in the collision group did not report their sex and gender and so the frequency count excluded them.

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Table 2

Variable	*M*	*SD* and (range)	1	2	3
1. CKI	20.06	3.47 (13 – 29)			
2. SOQ	104.75	14.29 (59 – 125)	-.08 [-.23, .07]		
Subscales					
Competitiveness	56.35	10.39 (28 – 65)			
Win Orientation	22.90	6.12 (12 – 30)			
Goal Orientation	25.52	4.83 (15 – 30)			
3. CAI	59.20	7.85 (36 – 75)	-.09 [-.24, .07]	-.01 [-.16, .15]	
4. Age	17.59	3.95 (14 – 32)	-.35** [-.48, -.20]	.24** [.09, .38]	.14 [-.01, .29]

Note. M and *SD* are used to represent mean and standard deviation, respectively. CKI = Concussion Knowledge Index; SOQ = Sport Orientation Questionnaire; CAI = Concussion Attitudes Index. Values in square brackets indicate the 95% confidence interval for each correlation. The confidence interval is a plausible range of population correlations that could have caused the sample correlation [40]. * indicates $p < .05$. ** indicates $p < .01$.

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Table 3

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	B (SE)	OR	97.5% CI
Past Nondisclosure (n = 156)[a]			
1. Age	.02 (2.20)	1.02	.92 – 1.12
2. Sex	.29 (.05)	1.34	.50 – 3.60
3. CKI	-.03(.01)	.95	.80 – 1.11
4. Rowing[b]	-2.05 (.82)*	.13*	.02 – 0.54
5. Soccer[b]	.28 (.72)	1.32	.32 – 5.51
6. SOQ	.03 (.01)*	1.03*	1.003 – 1.06
Nondisclosure Intention (n = 159)[c]			
1. Age	-.02 (.05)	.98	.89 – 1.07
2. Sex	.76 (.44)	2.13	.90 – 5.19
3. CKI	-.13 (.08)	.88	.74 – 1.02
4. Rowing[b]	-.15 (.53)	.86	.30 – 2.46
5. Soccer[b]	1.26 (.68)	3.51	.93 – 13.84
6. SOQ	.05 (.01)*	1.05*	1.03 – 1.08

Note. B= unstandardized beta; (SE)= standard error for the unstandardized beta; OR = Odds ratio; CI = Confidence Interval; CKI = Concussion Knowledge Index; SOQ = Sport Orientation Questionnaire. [a] Five participants removed in analyses due to missing data. [b] Rowing and soccer were compared to collision sports as the reference group. [c] Two participants removed from analyses due to missing data. * indicates $p < .05$.

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